Python & C Programming

Task 02 — Decision Making & Logic Building

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Part A: Even or Odd Checker

Takes a number as input and checks whether it is Even or Odd.

Python — even_odd.py

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is an Even Number")
else:
    print(f"{num} is an Odd Number")
```

Example Output:

```
Enter a number: 15
15 is an Odd Number
```

C — even_odd.c

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0)
        printf("%d is an Even Number\n", num);
    else
        printf("%d is an Odd Number\n", num);

    return 0;
}
```

Example Output:

```
Enter a number: 15
15 is an Odd Number
```

Part B: Largest of Three Numbers

Takes three numbers as input and finds the largest among them.

Python — largest_number.py

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

if a >= b and a >= c:
    largest = a
elif b >= a and b >= c:
```

```
    largest = b
else:
    largest = c

print(f"Largest Number = {largest}")
```

Example Output:

```
Enter first number: 10
Enter second number: 25
Enter third number: 18
Largest Number = 25.0
```

C — largest_number.c

```c
#include <stdio.h>

int main() {
    float a, b, c, largest;
    printf("Enter first number: ");  scanf("%f", &a);
    printf("Enter second number: "); scanf("%f", &b);
    printf("Enter third number: ");  scanf("%f", &c);

    if (a >= b && a >= c)
        largest = a;
    else if (b >= a && b >= c)
        largest = b;
    else
        largest = c;

    printf("Largest Number = %.1f\n", largest);
    return 0;
}
```

Example Output:

```
Enter first number: 10
Enter second number: 25
Enter third number: 18
Largest Number = 25.0
```

Part C: Student Grade Calculator

Accepts marks from the user and displays the grade based on the criteria below.

Marks	Grade
90 - 100	A
80 - 89	B
70 - 79	C
60 - 69	D
Below 60	F

Python — grade_calculator.py

```python
marks = float(input("Enter your marks: "))
```

```python
if 90 <= marks <= 100:
    grade = "A"
elif 80 <= marks < 90:
    grade = "B"
elif 70 <= marks < 80:
    grade = "C"
elif 60 <= marks < 70:
    grade = "D"
else:
    grade = "F"

print(f"Grade: {grade}")
```

Example Output:

```
Enter your marks: 85
Grade: B
```

C — grade_calculator.c

```c
#include <stdio.h>

int main() {
    float marks;
    char grade;

    printf("Enter your marks: ");
    scanf("%f", &marks);

    if (marks >= 90 && marks <= 100)      grade = 'A';
    else if (marks >= 80 && marks < 90)   grade = 'B';
    else if (marks >= 70 && marks < 80)   grade = 'C';
    else if (marks >= 60 && marks < 70)   grade = 'D';
    else                                   grade = 'F';

    printf("Grade: %c\n", grade);
    return 0;
}
```

Example Output:

```
Enter your marks: 85
Grade: B
```

Part D: Simple Login Validation

Stores a predefined username and password, then validates user input.

Python — login_validation.py

```python
stored_username = "admin"
stored_password = "password123"

username = input("Username: ")
password = input("Password: ")

if username == stored_username and password == stored_password:
    print("Login Successful")
else:
```

```
    print("Invalid Credentials")
```

Example Output (Success):

```
Username: admin
Password: password123
Login Successful
```

Example Output (Failure):

```
Username: admin
Password: wrongpass
Invalid Credentials
```

C — login_validation.c

```c
#include <stdio.h>
#include <string.h>

int main() {
    char stored_user[] = "admin";
    char stored_pass[] = "password123";
    char username[50], password[50];

    printf("Username: "); scanf("%s", username);
    printf("Password: "); scanf("%s", password);

    if (strcmp(username, stored_user) == 0 &&
        strcmp(password, stored_pass) == 0)
        printf("Login Successful\n");
    else
        printf("Invalid Credentials\n");

    return 0;
}
```

Example Output (Success):

```
Username: admin
Password: password123
Login Successful
```

Part E: Mini Cyber Security Challenge — Password Strength Checker

Checks password strength based on: length (min 8 chars), presence of at least one digit, and at least one uppercase letter.

Python — password_checker.py

```python
password = input("Enter Password: ")

has_length  = len(password) >= 8
has_digit   = any(char.isdigit() for char in password)
has_upper   = any(char.isupper() for char in password)

score = sum([has_length, has_digit, has_upper])

if score == 3:
    print("Strong Password")
elif score == 2:
```

```
    print("Moderate Password")
else:
    print("Weak Password")
```

Example Output:

```
Enter Password: Cyber123
Strong Password
```

C — password_checker.c

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char password[100];
    int has_length, has_digit = 0, has_upper = 0, score = 0;

    printf("Enter Password: ");
    scanf("%s", password);

    int len = strlen(password);
    has_length = (len >= 8);

    for (int i = 0; i < len; i++) {
        if (isdigit(password[i])) has_digit = 1;
        if (isupper(password[i])) has_upper = 1;
    }

    score = has_length + has_digit + has_upper;

    if (score == 3)      printf("Strong Password\n");
    else if (score == 2) printf("Moderate Password\n");
    else                 printf("Weak Password\n");

    return 0;
}
```

Example Output:

```
Enter Password: Cyber123
Strong Password
```

README.md

```
# Programming_Task_02_NawrinZawnaAN

## Part A - Even or Odd Checker (even_odd.py / even_odd.c)
Takes a number as input and checks if it is even or odd using the modulus operator.

## Part B - Largest of Three Numbers (largest_number.py / largest_number.c)
Takes three numbers and uses if-elif-else logic to find and display the largest.

## Part C - Student Grade Calculator (grade_calculator.py / grade_calculator.c)
Accepts marks and assigns a grade (A-F) based on predefined ranges using conditionals.

## Part D - Simple Login Validation (login_validation.py / login_validation.c)
```

Validates username and password against stored credentials using comparison operators.

Part E - Password Strength Checker (password_checker.py / password_checker.c)
Checks password strength based on length, digits, and uppercase letters.

How to Run Python Files
```
python even_odd.py
python largest_number.py
python grade_calculator.py
python login_validation.py
python password_checker.py
```

How to Compile & Run C Files
```
gcc even_odd.c -o even_odd && ./even_odd
gcc largest_number.c -o largest_number && ./largest_number
gcc grade_calculator.c -o grade_calculator && ./grade_calculator
gcc login_validation.c -o login_validation && ./login_validation
gcc password_checker.c -o password_checker && ./password_checker
```